

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

HR9TYuQR2LdcK22JxiNXzZGeST96UD3cYVGNdJGRhSmK

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Compounds: Cocaine, Codeine, Ecstasy

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Cocodecine

Formula: C₂₃H₂₈N₂O₄

Weight: 396.48

Drug-likeness: 0.72

Activity:

Neuromodulator

Target Analysis

Serotonin Transporter (SERT)

Binding Affinity: -11.3

Involved in mood regulation and implicated in depression and anxiety disorders.

Norepinephrine Transporter (NET)

Binding Affinity: -10.5

Important in the modulation of attention and response to stress.

Dopamine Transporter (DAT)

Binding Affinity: -10.9

Crucial for reward-based behavior and implicated in disorders such as ADHD and Parkinson's disease.

Toxicity Profile

Risk Level: High

Affected Systems: Heart, Brain, Liver

Mitigation Suggestions:

- Use of beta-blockers
- Monitoring of liver enzymes
- Cognitive behavioral therapy

Pharmacokinetics

Absorption:

Rapid oral absorption with high bioavailability

Distribution:

Widely distributed in the central nervous system and adipose tissues

Metabolism:

Primarily hepatic through cytochrome P450 enzymes

Excretion:

Renal

Half-life: Approx. 8-12 hours

Detailed Analysis

Cocodectine is a hypothetical compound synthesized by combining unique active moieties from cocaine, codeine, and ecstasy to leverage distinct pharmacological profiles into a single agent. It acts on multiple neurotransmitter pathways, primarily modulating serotonin, norepinephrine, and dopamine reuptake, which coalesce to potentiate neuromodulation and mood stabilization. However, these synergistic effects come with high toxicity risks, including cardiotoxic and neurotoxic potentials. The compound offers promising applications for mood disorders and pain management, but the complexity of its side effects necessitates careful progression through preclinical and clinical stages. Initial assessments indicate fast absorption and wide distribution, with hepatic metabolism and renal excretion, offering a moderate half-life conducive to therapeutic settings. Further research focused on selectivity and long-term safety could enhance viability significantly.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>